

Indian Statistical Institute
Mid Semestral Examination: 2024-25

Course Name: M. Math 1st year

Subject Name : Analysis of several variables

Date: 10.09.2022, Maximum Marks: 30, Duration: Two and a half hours

You are allowed to consult lecture notes.

Answer all questions

- ✓ 1. Compute the derivative of the function $f : M_n(\mathbb{R}) \rightarrow M_n(\mathbb{R})$ defined by $f(A) = A^3$.
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- ✓ 2. Consider the system of equations:

$$2x^2 - x^2z^2 - y^2 = 0, xyz = 1.$$

Prove that there exists a neighborhood of the point $(1, 1, 1)$ in which the set of solutions to this system of equations can be described as a graph of a function of z defined on an open set containing 1. 5

- ✓ 3. Using the method of Lagrange multipliers, find the maximum value of the function $g(x, y, z) = 2x + 2y - z$ on the set $x^2 + y^2 + z^2 = 2$. 5

- ✓ 4. Suppose U is an open convex set in $\mathbb{R}^n$ and $f : U \rightarrow \mathbb{R}$ be a C^3 -function. Prove that for all a in U and h in $\mathbb{R}^n$ such that $\|h\|$ is sufficiently small,

$$f(a+h) = f(a) + \langle \nabla f(a), h \rangle + \frac{1}{2} D^2 f(a; h) + \frac{1}{6} D^3 f(a; h) + \|h\|^3 E(h),$$

where E is a real-valued function defined on an open set containing zero such that $|E(h)| \rightarrow 0$ as $\|h\| \rightarrow 0$. 5

- ✓ 5. Recall that the derivative of the function $\det : GL_n(\mathbb{R}) \rightarrow \mathbb{R}$ is given by $D(\det)(A)(X) = \det(A) \text{Tr}(A^{-1}X)$.

- ✓ (a) Compute the dimension of the vector space $\text{Ker}(D(\det)(I))$, where I denotes the identity matrix in $M_n(\mathbb{R})$.

- ✓ (b) Show that $\text{SL}_n(\mathbb{R}) := \{A \in M_n(\mathbb{R}) : \det(A) = 1\}$ is a regular $n^2 - 1$ -level surface in $\mathbb{R}^{n^2}$. 2 + 3 = 5

- ✓ 6. Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is C^1 . Prove that f is not one-one. 5

(Hint: If $Df(x, y) = 0$ for all x, y , then f is constant and hence not one-one. So, without loss of generality, suppose that there exists $(x_0, y_0) \in \mathbb{R}^2$ such that $\frac{\partial f}{\partial y}(x_0, y_0) \neq 0$. Then you can consider the equation

$$h : \mathbb{R}^2 \rightarrow \mathbb{R}, h(x, y) = f(x, y) - f(x_0, y_0).$$

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